

Problem-Solving & Data



Ratios and proportions

- 1 A recipe calls for 3 cups of flour for every 2 cups of sugar. If a baker uses 9 cups of flour, how many cups of sugar are needed?
 - 2 On a map, 1 inch represents 25 miles. If two cities are 7 inches apart on the map, what is the actual distance between them?
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Percent problems

- 3 A shirt originally priced at 40 dollars is on sale for 25 percent off. Find the sale price.
 - 4 A population increased from 4000 to 4600. Find the percent increase.
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Unit conversions and rates

- 5 A car travels at 60 miles per hour. How many miles does it travel in 45 minutes?
 - 6 A factory produces 480 units in an 8-hour shift. Find the production rate in units per minute.
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Reading and interpreting scatterplots and tables

- 7 A table shows study hours and test scores: (2,65), (4,75), (6,85), (8,95). Describe the relationship between study hours and test score, and predict the score for 5 hours of study.
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Measures of center: mean, median, and mode

- 8 Find the mean, median, and mode of the data set: 4, 7, 7, 9, 12, 15.
 - 9 A data set has mean 20 and 5 values. If four of the values are 15, 18, 22, 25, find the fifth value.
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Measures of spread and the effect of outliers

- 10 Find the range and explain the effect on the mean if an outlier of 50 is added to the data set 10, 12, 14, 16, 18.
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Probability from data tables

- 11 A survey of 200 students found 120 prefer math and 80 prefer science. If a student is selected at random, find the probability the student prefers math.
- 12 A two-way table shows: 50 students play sports and pass an exam; 20 play sports and fail; 30 don't play sports and pass; 10 don't play sports and fail. Find the probability a randomly selected student plays sports, given the student passed the exam.
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Population, sample, and margin of error

- 13 A poll of 500 randomly selected voters found that 55% support a proposal, with a margin of error of 4%. Interpret this result and state the range of plausible values for the true population percentage.
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Weighted averages

- 14 A student's final grade is weighted: homework counts 20%, tests count 50%, and the final exam counts 30%. If the student scores 90 on homework, 80 on tests, and 85 on the final exam, find the weighted final grade.
- 15 A store mixes 3 pounds of coffee costing 8 dollars per pound with 2 pounds of coffee costing 13 dollars per pound. Find the price per pound of the mixture.
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Multi-step unit conversion problems

- 16 A machine produces 15 items per minute. Find the number of items produced in 3.5 hours.
- 17 A car uses fuel at a rate of 28 miles per gallon. If gas costs 3.50 dollars per gallon, find the fuel cost for a 420-mile trip.
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Percent change applied twice (successive changes)

- 18 A stock price increases by 20% one year, then decreases by 20% the next year. If the original price was 100 dollars, find the final price, and explain why it is not back to 100 dollars.

Density and rate-based problem solving

- 19** A population density is 150 people per square mile. If a city has an area of 24 square miles, find the total population.
- 20** Three workers can complete a job in 12 hours together. If they work at the same combined rate, how many hours would it take 2 workers with the same total capacity to complete a job that is 1.5 times as large?